

KIRKKONUMMI MAKILUOTO, Finland

WMO: 027940

Lat: 59.920N

Lon: 24.349E

Elev: 2

StdP: 101.30

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.2	-13.0	-18.2	0.8	-15.8	-15.2	1.0	-11.9	18.4	1.7	17.4	1.2	7.2	30	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	23.9	20.4	22.4	19.5	21.1	18.5	21.2	23.3	20.1	22.2	18.8	20.7	5.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.0	22.8	19.3	14.0	21.8	17.9	12.9	20.3	61.4	23.5	57.8	22.2	53.6	20.9	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.8	15.2	13.7	-16.6	25.3	4.5	2.5	-19.8	27.1	-22.5	28.6	-25.0	30.0	-28.3	31.8		
			-16.8	21.4	4.7	1.6	-20.2	22.5	-22.9	23.5	-25.6	24.3	-29.0	25.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-3.0	-2.8	-0.7	3.5	9.1	13.7	17.8	17.2	13.4	7.6	4.0	1.0
	DBStd	7.93	4.54	5.07	3.32	2.13	3.09	2.23	2.71	2.27	2.53	2.99	3.46	3.97
	HDD10.0	1895	403	359	333	196	54	2	0	0	6	83	181	279
	HDD18.3	4258	661	593	591	446	286	139	44	48	149	332	431	538
	CDD10.0	720	0	0	0	0	27	112	242	224	106	9	0	0
	CDD18.3	41	0	0	0	0	0	0	27	14	0	0	0	0
	CDH23.3	69	0	0	0	0	0	1	54	14	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	1	0	0	0	0
Wind	WSAvg	7.3	8.0	7.6	7.4	6.4	6.2	6.4	5.8	6.3	7.1	8.1	8.7	9.3
Precipitation	PrecAvg	579	46	34	29	27	30	49	59	73	51	67	64	49
	PrecMax	764	74	94	54	59	73	87	141	153	90	109	155	83
	PrecMin	439	5	1	6	5	3	14	4	8	18	33	10	13
	PrecStd	74	17	24	12	16	22	19	35	34	23	21	34	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	4.0	6.1	11.0	19.1	21.0	26.6	24.8	19.4	14.1	10.0	7.9
		MCWB	3.9	2.5	3.2	7.3	13.4	16.1	21.7	21.0	16.4	12.2	9.3	7.3
	2%	DB	4.2	3.2	4.6	9.0	17.0	19.4	24.8	23.2	18.3	13.0	9.3	7.1
		MCWB	3.5	2.3	2.7	6.2	12.8	15.7	20.9	20.0	15.8	11.7	8.6	6.5
	5%	DB	3.4	2.6	3.5	7.7	15.1	18.2	23.4	21.8	17.4	12.3	8.6	6.2
		MCWB	2.8	2.0	2.2	5.3	11.8	15.3	20.2	18.9	15.4	11.1	7.9	5.5
	10%	DB	2.3	2.0	2.7	6.5	13.6	17.1	22.1	20.6	16.6	11.5	7.9	5.4
		MCWB	1.8	1.5	1.7	4.6	10.8	14.5	19.4	18.0	14.9	10.4	7.2	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.2	4.0	7.6	14.5	17.7	22.6	21.8	17.2	13.0	9.6	7.6
		MCDB	4.5	3.5	5.5	10.4	18.1	19.5	25.3	24.0	18.6	13.7	9.9	7.7
	2%	WB	3.6	2.6	3.1	6.6	13.4	16.7	21.7	20.8	16.4	12.2	8.9	6.6
		MCDB	4.1	3.0	4.0	8.6	16.3	18.7	23.8	22.7	17.6	12.7	9.2	6.9
	5%	WB	2.9	2.1	2.5	5.7	12.4	15.8	20.7	19.6	15.8	11.5	8.1	5.6
		MCDB	3.4	2.4	3.3	7.4	14.8	17.7	23.0	21.4	16.9	12.1	8.5	6.1
	10%	WB	2.0	1.6	2.0	4.8	11.2	14.9	19.9	18.4	15.2	10.6	7.4	4.8
		MCDB	2.4	1.9	2.6	6.2	13.3	16.8	21.9	20.1	16.4	11.4	7.8	5.4
Mean Daily Temperature Range	5% DB	MDBR	3.8	3.4	4.0	4.0	4.8	5.0	5.2	4.8	4.2	3.7	2.8	3.2
		MCDBR	2.8	2.5	3.9	5.6	6.8	6.3	6.3	5.7	5.0	3.8	2.9	3.0
		MCWBR	2.8	2.3	2.8	3.5	3.9	3.7	3.6	3.6	3.4	3.1	3.1	3.1
	5% WB	MCDBR	2.9	2.4	3.5	5.0	6.1	5.6	5.4	4.8	4.4	3.4	2.8	2.9
		MCWBR	2.9	2.3	2.7	3.3	4.0	3.6	3.3	3.3	3.2	3.2	3.0	3.1
Clear-Sky Solar Irradiance	taub	0.279	0.312	0.334	0.365	0.362	0.359	0.384	0.378	0.349	0.335	0.309	0.264	
	taud	2.142	2.194	2.255	2.317	2.402	2.474	2.429	2.444	2.511	2.400	2.221	2.087	
	Ebn at Noon	483	675	790	833	866	874	839	815	783	658	444	346	
	Edh at Noon	44	73	94	106	104	99	102	93	74	60	42	31	
All-Sky Solar Radiation	RadAvg	0.25	0.81	2.31	3.87	5.35	5.58	5.35	4.17	2.54	1.10	0.30	0.15	
	RadStd	0.04	0.14	0.29	0.41	0.57	0.41	0.47	0.42	0.33	0.16	0.05	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-134	N/A	N/A	N/A

Nomenclature: See separate page